

Ch 7: How do Organisms Reproduce

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Why is reproduction necessary for the continuity of a species? What would happen if organisms did not reproduce? (PYQ 2019)

Q2. Define reproduction. Name the two main types of reproduction found in organisms. (PYQ 2020)

Q3. What is the basic event in reproduction? Why is DNA copying important during reproduction? (PYQ 2022)

Q4. Why is DNA copying not perfectly accurate? State one advantage of variations produced during reproduction. (PYQ 2019)

Q5. What is asexual reproduction? Name any two methods of asexual reproduction with one example each. (PYQ 2020)

Q6. What is binary fission? Name one organism that reproduces by binary fission. (PYQ 2022)

Q7. Differentiate between binary fission and multiple fission with one example each. (PYQ 2019)

Q8. What is fragmentation? Why is fragmentation not possible in all multicellular organisms? (PYQ 2020)

Q9. What is regeneration? Name one organism that shows regeneration. Is regeneration same as reproduction? (PYQ 2022)

Q10. What is budding? Explain budding in Hydra in two steps. (PYQ 2019)

Q11. What is vegetative propagation? Give two examples of plants that reproduce by vegetative propagation. (PYQ 2020)

Q12. State two advantages of vegetative propagation in agriculture. (PYQ 2022)

Q13. What is spore formation? Name one organism that reproduces by spores. (PYQ 2019)

Q14. Why are spores protected by thick walls? How does this help the organism? (PYQ 2020)

Q15. What is tissue culture? Why is it useful in producing disease-free plants? (PYQ 2024)

Q16. What is sexual reproduction? How is it different from asexual reproduction? (PYQ 2022)

Q17. Why is sexual reproduction considered a source of variation? (PYQ 2019)

Q18. What is the function of flowers in plants? Name the male and female reproductive parts of a flower. (PYQ 2020)

Q19. Name the parts of the stamen and carpel. What does each part produce? (PYQ 2022)

Q20. What is pollination? Name the two types of pollination. (PYQ 2019)

Q21. Differentiate between self-pollination and cross-pollination. (PYQ 2020)

Q22. What is fertilisation in flowering plants? Where does it occur? (PYQ 2022)

Q23. What is a pollen tube? Why is it formed after pollination? (PYQ 2019)

Q24. What changes occur in the ovule and ovary after fertilisation? (PYQ 2020)

Q25. Define germination. Name two conditions necessary for seed germination. (PYQ 2022)

Q26. What is puberty? Name two changes that occur in boys and girls during puberty. (PYQ 2019)

Q27. Name the male reproductive organs in humans. What is the function of testes? (PYQ 2020)

Q28. What is the role of testosterone in the male reproductive system? (PYQ 2022)

Q29. Why are testes located outside the abdominal cavity in the scrotum? (PYQ 2019)

Q30. What is semen? Name any two glands that add fluids to sperms. (PYQ 2020)

Q31. Name the female reproductive organs in humans. What is the function of ovaries? (PYQ 2022)

Q32. What is ovulation? Name the hormone mainly associated with female secondary sexual characters. (PYQ 2019)

Q33. What is fertilisation in humans? Where does fertilisation usually take place? (PYQ 2020)

Q34. What is implantation? Where does the embryo get implanted? (PYQ 2022)

Q35. What is placenta? State two functions of placenta. (PYQ 2019)

Q36. What is menstruation? Why does menstruation occur if fertilisation does not take place? (PYQ 2020)

Q37. What are contraceptive methods? State any two reasons for using contraception. (PYQ 2022)

Q38. Name any two mechanical methods of contraception. How do they prevent pregnancy? (PYQ 2019)

Q39. What are sexually transmitted infections? Name any two examples. (PYQ 2020)

Q40. Why is reproductive health important? Mention any two practices for maintaining reproductive health. (PYQ 2024)

SECTION B - 3 MARKS

Q1. Explain why DNA copying is essential in reproduction. How do minor errors in DNA copying help a species survive changing environments? (PYQ 2019)

Q2. Compare asexual and sexual reproduction: (PYQ 2020)

Basis Asexual Reproduction Sexual Reproduction

Number of parents

Gamete formation

Variation produced

Speed of process

Example

Q3. Explain binary fission in Amoeba and multiple fission in Plasmodium. State one difference between the two. (PYQ 2022)

Q4. Describe budding in Hydra with the help of steps. Why is the bud genetically similar to the parent? (PYQ 2019)

Q5. Explain fragmentation in Spirogyra. Why can fragmentation occur only in simple multicellular organisms? (PYQ 2020)

Q6. What is regeneration? Explain regeneration in Planaria. Why is regeneration not the same as reproduction in all cases? (PYQ 2022)

Q7. Describe spore formation in Rhizopus. Mention the role of sporangium, spores, and thick spore walls. (PYQ 2019)

Q8. Explain vegetative propagation with examples of potato, ginger, Bryophyllum, and money plant. (PYQ 2020)

Q9. State three advantages of vegetative propagation for farmers and gardeners. (PYQ 2022)

Q10. Compare natural vegetative propagation and artificial vegetative propagation with examples. (PYQ 2019)

Q11. What is tissue culture? Explain how many plants can be produced from a small piece of plant tissue under sterile conditions. (PYQ 2024)

Q12. Explain why sexual reproduction gives rise to more variation than asexual reproduction. How is this useful for evolution? (PYQ 2020)

Q13. Draw a labelled diagram of a flower and mark: sepals, petals, stamen, anther, filament, carpel, stigma, style, ovary, and ovule. State the function of stamen and carpel. (PYQ 2022)

Q14. Explain the structure of the stamen and carpel. Mention what is produced in the anther and ovule. (PYQ 2019)

Q15. Compare self-pollination and cross-pollination: (PYQ 2020)

Basis Self-Pollination Cross-Pollination

Source of pollen

Need of pollinating agent

Variation produced

Example

Advantage

Q16. Explain the events that take place in a flower after pollination. Include pollen tube formation, fertilisation, and seed formation. (PYQ 2022)

Q17. What changes occur in different parts of a flower after fertilisation? Explain the fate of ovary, ovule, petals, sepals, and zygote. (PYQ 2019)

Q18. Describe the process of seed germination. What conditions are required for a seed to germinate? (PYQ 2020)

Q19. Explain the changes that occur in boys and girls at puberty. Why are these changes called secondary sexual characters? (PYQ 2022)

Q20. Describe the male reproductive system in humans. Mention the role of testes, vas deferens, prostate gland, seminal vesicle, and urethra. (PYQ 2019)

Q21. Describe the female reproductive system in humans. Mention the role of ovaries, oviducts, uterus, cervix, and vagina. (PYQ 2020)

Q22. Explain the pathway of sperm from its formation to release from the body. Name the glands that add secretions to sperms. (PYQ 2022)

Q23. Explain ovulation, fertilisation, implantation, and pregnancy in correct sequence. (PYQ 2019)

Q24. What is placenta? Explain how placenta helps in nutrition, respiration, and excretion of the developing embryo. (PYQ 2020)

Q25. Explain the menstrual cycle in simple terms. What happens to the uterine lining if fertilisation occurs and if fertilisation does not occur? (PYQ 2022)

Q26. Compare male and female reproductive systems: (PYQ 2019)

Basis	Male Reproductive System	Female Reproductive System
Gamete produced		
Main organ		
Hormone		
Site of gamete formation		
Site of fertilisation		

Q27. What are the different methods of contraception? Explain barrier methods, chemical methods, intrauterine devices, and surgical methods. (PYQ 2020)

Q28. Why is family planning important? Explain how contraception helps in reproductive health and population control. (PYQ 2022)

Q29. What are sexually transmitted infections? How can they be prevented? Name any two bacterial or viral STIs. (PYQ 2019)

Q30. Distinguish between fertilisation and germination, pollination and fertilisation, and embryo and zygote. (PYQ 2020)

Q31. What happens when: (i) pollen does not reach stigma (ii) fallopian tube is blocked (iii) uterus lining is not maintained after ovulation? (PYQ 2022)

Q32. Explain why reproductive health education is important for adolescents. Mention hygiene, awareness, prevention of infections, and responsible decision-making. (PYQ 2024)

Q33. Describe how reproduction in plants and humans shows the same basic idea of gamete formation, fertilisation, zygote formation, and development. (PYQ 2019)

Q34. Write the differences between external fertilisation and internal fertilisation with examples. (PYQ 2020)

Q35. Explain why a child receives genetic material from both parents in sexual reproduction. How does this create variation? (PYQ 2022)

SECTION C - 5 MARKS

Q1. Explain the importance of reproduction in organisms. Describe how DNA copying produces similar but not identical individuals, and explain why variation is useful for the survival of a species. (PYQ 2022)

Q2. Compare asexual reproduction and sexual reproduction in detail across five parameters. Give two examples of each and explain why sexual reproduction produces greater variation. (PYQ 2020)

Q3. Describe the different methods of asexual reproduction in organisms: binary fission, multiple fission, fragmentation, regeneration, budding, vegetative propagation, and spore formation. Give one example for each. (PYQ 2019)

Q4. Draw labelled diagrams and explain the following modes of asexual reproduction: (i) binary fission in Amoeba (ii) budding in Hydra (iii) spore formation in Rhizopus. (PYQ 2022)

Q5. Explain vegetative propagation in plants. Describe natural methods using potato, ginger, Bryophyllum, and money plant, and artificial methods such as cutting, grafting, layering, and tissue culture. State advantages. (PYQ 2020)

Q6. What is regeneration? Explain regeneration in Planaria. Compare regeneration, fragmentation, and budding with respect to body organisation, process, and examples. (PYQ 2019)

Q7. Describe sexual reproduction in flowering plants with a labelled diagram of a flower. Explain the functions of sepals, petals, stamens, carpel, stigma, style, ovary, and ovule. (PYQ 2022)

Q8. Explain pollination and fertilisation in flowering plants. Distinguish between self-pollination and cross-pollination. Explain pollen tube formation and the fate of ovary and ovule after fertilisation. (PYQ 2020)

Q9. Draw a labelled diagram of the longitudinal section of a flower. Trace the path of the male gamete from anther to ovule and explain how zygote, embryo, seed, and fruit are formed. (PYQ 2019)

Q10. Write a comprehensive note on reproduction in plants covering: asexual reproduction, vegetative propagation, flower structure, pollination, fertilisation, seed formation, and germination. (PYQ 2022 comprehensive)

Q11. Describe the changes that occur during puberty in boys and girls. Explain the role of sex hormones and why the body becomes capable of reproduction only after puberty. (PYQ 2020)

Q12. Draw a labelled diagram of the male reproductive system in humans. Explain the functions of testes, scrotum, vas deferens, seminal vesicle, prostate gland, urethra, and penis. (PYQ 2019)

Q13. Draw a labelled diagram of the female reproductive system in humans. Explain the functions of ovaries, oviducts, uterus, cervix, and vagina. Mention the site of fertilisation and implantation. (PYQ 2022)

Q14. Explain the complete process of human reproduction from gamete formation to fertilisation, implantation, placenta formation, development of embryo, and birth. (PYQ 2020)

Q15. What is menstruation? Explain the changes in the uterus during the menstrual cycle. What happens if fertilisation takes place and what happens if fertilisation does not take place? (PYQ 2019)

Q16. Explain the structure and function of the placenta. How does it help in exchange of nutrients, oxygen, carbon dioxide, and wastes between mother and developing embryo? (PYQ 2022)

Q17. Compare reproduction in plants and humans across five parameters: reproductive organs, gametes, fertilisation site, zygote development, and protection/nutrition of embryo. (PYQ 2020)

Q18. What are contraceptive methods? Explain barrier methods, chemical methods, intrauterine devices, and surgical methods. Mention their role in preventing pregnancy and reducing the spread of infections. (PYQ 2019)

Q19. Write a detailed note on reproductive health. Include personal hygiene, adolescent awareness, contraception, family planning, prevention of sexually transmitted infections, and responsible health practices. (PYQ 2024)

Q20. Case Study: A flower is pollinated and the pollen grain germinates on the stigma. The pollen tube grows through the style and reaches the ovule. (PYQ 2024 case-based)

Question Answer

What is pollination?

Why does the pollen tube grow towards the ovule?

Where does fertilisation occur in the flower?

What does the ovule become after fertilisation?

What does the ovary become after fertilisation?

Q21. Case Study: A couple wants to avoid pregnancy for some time and also wants protection from infections. A doctor explains different contraceptive methods. (PYQ 2024 case-based)

Question Answer

Which contraceptive method can reduce the risk of infections?

Name one chemical method of contraception.

What is an IUCD?

What are surgical contraceptive methods called in males and females?

Why should contraception be used responsibly under proper guidance?

Q22. Complete the following comparison table related to asexual reproduction: (PYQ 2022)

Mode Organism Main Feature Advantage

Binary fission

Multiple fission

Fragmentation

Budding

Spore formation

Vegetative propagation

Q23. Complete the following comparison table related to human reproductive organs: (PYQ 2020)

Organ System Function Important Point

Testes Male

Scrotum Male

Vas deferens Male

Ovaries Female

Oviduct Female

Uterus Female

Placenta Female

Q24. Answer the following questions related to reproduction: (PYQ 2023)

Question Answer

Why is variation important for a species?

Why are testes present in the scrotum?

Why is the uterus lining thickened every month?

Why are spores useful for survival in unfavourable conditions?

Why is cross-pollination generally better for variation?

Q25. Distinguish between the following pairs with examples wherever required: (i) pollination and fertilisation (ii) zygote and embryo (iii) fission and budding (iv) menstruation and pregnancy (v) internal and external fertilisation. (PYQ 2022)

Q26. Write a complete centum-level answer on the chapter How do Organisms Reproduce covering: need for reproduction, DNA copying and variation, asexual methods, sexual reproduction in plants, human reproductive systems, fertilisation, placenta, menstruation, contraception, and reproductive health. (PYQ 2022 comprehensive)